TASK WK1-PM1 | WEEK 1, PROJECT MODULE 1

Cybersecurity Testing Lab Setup

VMware Workstation Pro and Kali Linux on an isolated 10.0.0.0/24 NAT network

Item	Details
Course	Ethical Hacking and Cybersecurity, Networkwalks Academy
Report type	Practical lab setup report
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Contents

1. Objective and Requirements **1**

2. Lab Environment **1**

2.1 Host and virtual machine **1**

2.2 Addressing plan **2**

3. Software Installation (Steps 1 and 2) **2**

4. NAT Network Configuration (Step 3) **2**

5. Kali Linux VM Configuration (Step 4) **3**

5.1 Network adapter **3**

5.2 Clipboard and drag and drop **4**

5.3 Shared folder **4**

5.4 Kali Linux running **5**

6. Static IP Configuration (Step 5) **5**

6.1 Connectivity tests **6**

7. Snapshot (Step 6) **7**

8. Verification and Troubleshooting **7**

8.1 Verification checklist **7**

8.2 If Kali has no Internet **7**

9. Conclusion **8**

1. Objective and Requirements

This report documents the setup of a cybersecurity testing lab. Kali Linux acts as the attacking machine on an isolated NAT network (10.0.0.0/24), giving a safe, contained environment for ethical hacking practice. The task names VirtualBox as the base; **VMware Workstation Pro** was used instead, with its VMware equivalent for each requirement. The table below records what the screenshots in this report show.

#	Requirement	Result shown in the screenshots	Evidence
1	Hypervisor as the base	VMware Workstation Pro 26H1	Figure 1
2	Kali Linux as attacking machine	Kali Linux VM running in VMware	Figure 6
3	Subnet 10.0.0.0/24	VMnet8 subnet 10.0.0.0, mask 255.255.255.0	Figure 2
4	NAT network on 10.0.0.0/24	VMnet8 (NAT) attached to the Kali VM	Figures 2, 3
5	Clipboard and drag/drop enabled	Guest Isolation options ticked	Figure 4
6	Shared folder (Downloads) from host	Folder sharing set to Always enabled; folder list still empty	Figure 5
7	Kali IP address 10.0.0.2/24	Manual IPv4 set to 10.0.0.224/24; terminal shows 10.0.0.100/24	Figures 7, 8
8	Full Internet access	Pings to 8.8.8.8 and google.com succeed (0% loss)	Figure 8

2. Lab Environment

All lab machines share one custom VMware NAT network (VMnet8). They can reach each other and the Internet, but stay isolated from the physical LAN.

2.1 Host and virtual machine

Item	Value
Host operating system	Windows 11 Pro, 64-bit (build 26300)
Host memory	11,932 MB (about 12 GB)
Hypervisor	VMware Workstation Pro 26H1 (version 26.0.0.25388281)
Guest operating system	Kali Linux (attacking machine)
VM resources	4 GB memory, 2 processors, 60 GB disk
VM network	Custom: VMnet8 (NAT), subnet 10.0.0.0/24

2.2 Addressing plan

Machine	Role	IP address	Phase
Kali Linux	Attacker	10.0.0.224/24 (as configured)	1 (this report)
Windows 7	Target (optional)	10.0.0.7/24	2
Android 9x	Target (optional)	10.0.0.9/24	2
Windows 10	Target (optional)	10.0.0.10/24	2
Windows 11	Target (optional)	10.0.0.11/24	2
Windows Server 2016	Target (optional)	10.0.0.16/24	2

The NAT gateway is `10.0.0.254` and DHCP hands out `10.0.0.100` to `10.0.0.200`. Fixed lab addresses outside that range never collide with DHCP leases.

3. Software Installation (Steps 1 and 2)

7-Zip extracts compressed virtual machine images, and VMware Workstation Pro is the hypervisor that runs the lab.

1. Download and install 7-Zip from 7-zip.org.
2. Download and install VMware Workstation Pro from VMware's official download page.
3. Open VMware and check **Help > About** to confirm the installed version (Figure 1).

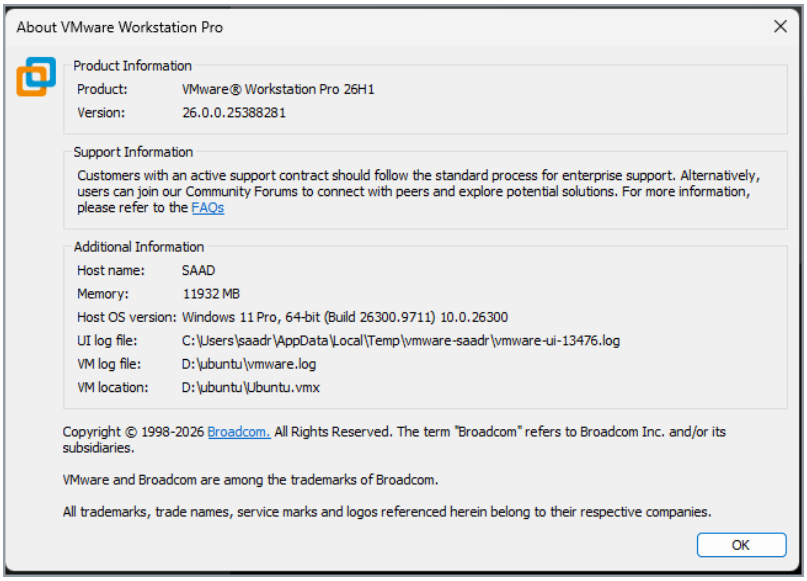


Figure 1. *VMware Workstation Pro 26H1 installed on the Windows 11 Pro host.*

4. NAT Network Configuration (Step 3)

A custom NAT network gives every VM a private address range with shared Internet access through the host.

1. Open **Edit > Virtual Network Editor** and click **Change Settings** (administrator approval).
2. Select **VMnet8**, whose type is **NAT**.

3. Set **Subnet IP** to `10.0.0.0` and **Subnet mask** to `255.255.255.0`.
4. Click **NAT Settings** and set **Gateway IP** to `10.0.0.254`. VMware's default gateway ends in .2, so a different gateway address is used to keep 10.0.0.2 free for Kali.
5. Click **DHCP Settings** and set the range to `10.0.0.100` to `10.0.0.200`. Click **Apply**, then **OK**.

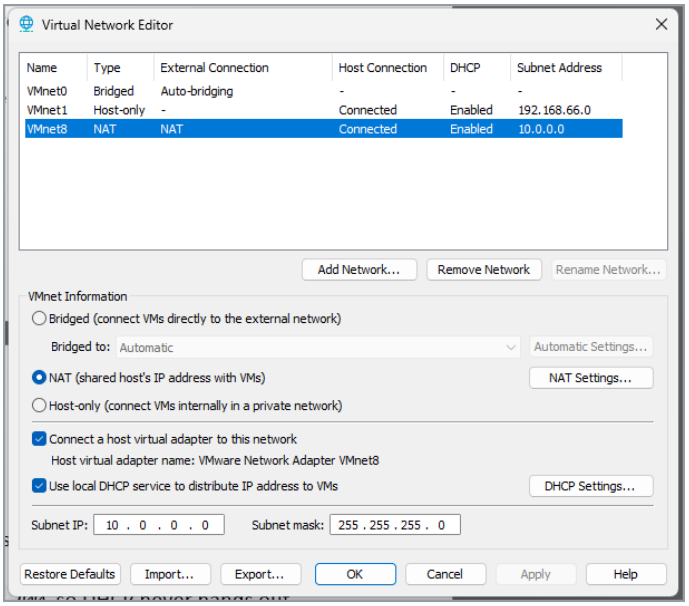


Figure 2. *Virtual Network Editor showing VMnet8 (NAT) with subnet 10.0.0.0 and mask 255.255.255.0.*

5. Kali Linux VM Configuration (Step 4)

Kali Linux was already installed in VMware, so the download and import steps were skipped. The VM was then configured to meet the lab requirements.

5.1 Network adapter

In **VM > Settings > Hardware > Network Adapter**, the adapter is set to **Custom: VMnet8 (NAT)**, with **Connect at power on** ticked.

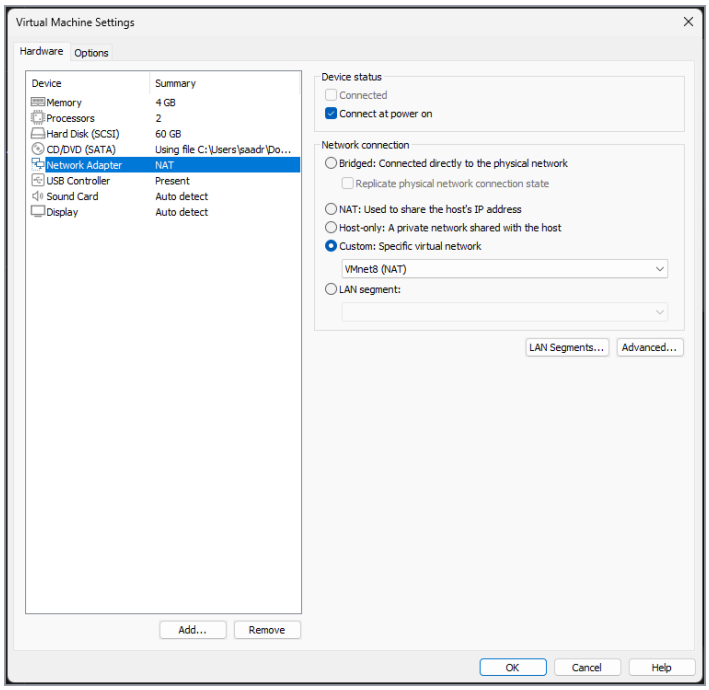


Figure 3. *Network Adapter set to the custom VMnet8 (NAT) network.*

5.2 Clipboard and drag and drop

In **VM > Settings > Options > Guest Isolation**, both **Enable drag and drop** and **Enable copy and paste** are ticked. This relies on VMware Tools (`open-vm-tools-desktop`) running inside Kali.

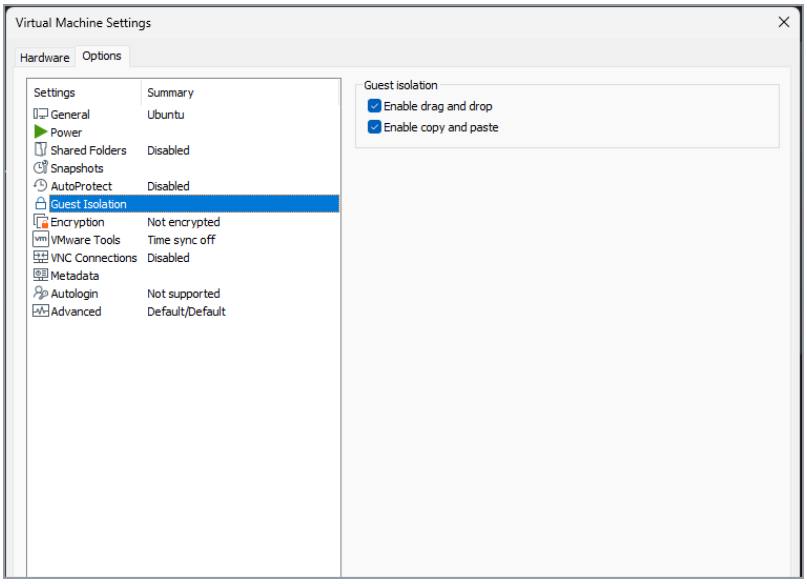


Figure 4. *Guest Isolation options enabled for drag and drop and copy and paste.*

5.3 Shared folder

In **VM > Settings > Options > Shared Folders**, folder sharing is set to **Always enabled**. Host folders are added with **Add**, and inside Kali a share appears under `/mnt/hgfs/`. At the time of the screenshot the folder list is still empty, so the Downloads folder has not been added yet.

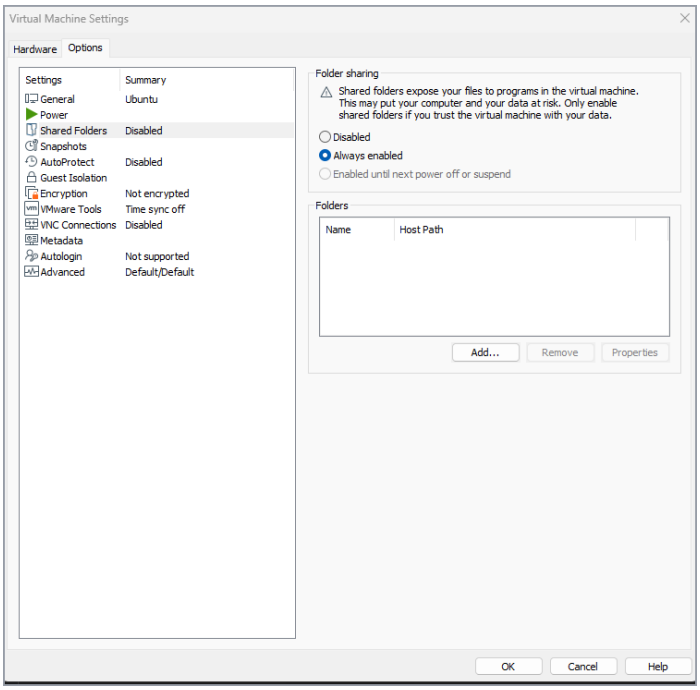


Figure 5. *Shared Folders settings with folder sharing set to Always enabled and an empty folder list.*

5.4 Kali Linux running



Figure 6. *Kali Linux desktop running inside VMware Workstation Pro.*

6. Static IP Configuration (Step 5)

Kali uses a manual IPv4 address on the lab network. The task asks for 10.0.0.2/24; the configuration captured in Figure 7 uses 10.0.0.224/24 with the same gateway.

1. Right-click the network icon in the top panel and choose **Edit Connections**.
2. Open **Wired connection 1** and select the **IPv4 Settings** tab.
3. Set **Method** to `Manual`.
4. Add the address (`10.0.0.224` in Figure 7), netmask 24 and gateway `10.0.0.254`.
5. Set the DNS server to `8.8.8.8` (use `10.0.0.254` if the Internet gives issues), then click **Save**.

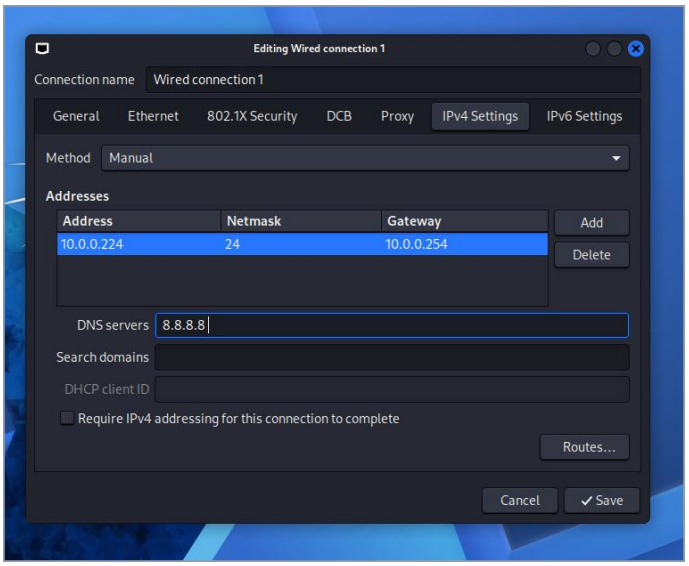


Figure 7. *IPv4 settings of Wired connection 1 in Kali Linux.*

6.1 Connectivity tests

Connectivity was tested from a terminal in Kali:

```
ip a
ping -c 4 10.0.0.254
ping -c 4 8.8.8.8
ping -c 4 google.com
```

In Figure 8, `eth0` shows `10.0.0.100/24` marked dynamic, a DHCP lease from the VMnet8 range. All three ping tests succeed with 0% packet loss, so the VM reaches the gateway, the Internet and DNS.

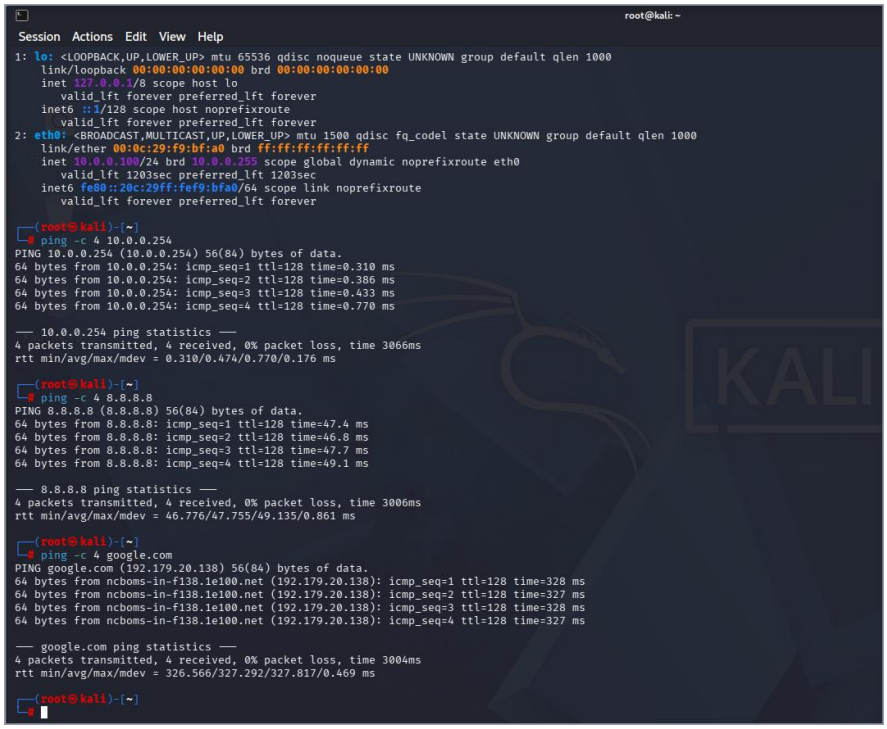


Figure 8. *Output of ip a and ping tests to the gateway, 8.8.8.8 and google.com.*

7. Snapshot (Step 6)

A snapshot of the working Kali VM allows the lab to be rolled back after experiments or mistakes. The **Take Snapshot** dialog (VM > Snapshot > Take Snapshot) was opened with the default name `Snapshot 1` and an empty description (Figure 9).

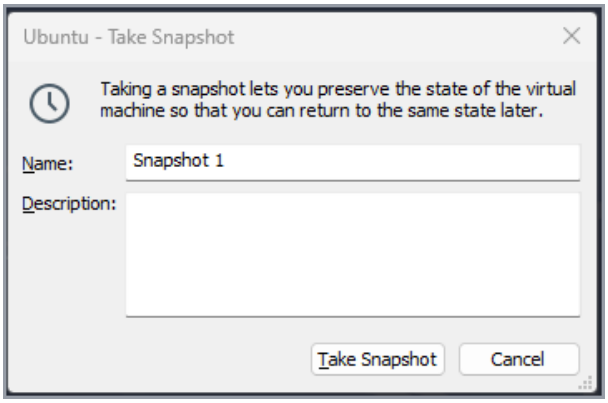


Figure 9. *Take Snapshot dialog for the Kali VM, with the default name Snapshot 1.*

8. Verification and Troubleshooting

8.1 Verification checklist

Check	Command or location	Expected result
IP address	`ip a`	An address from 10.0.0.0/24 on eth0 (10.0.0.2 as required)
Gateway reachable	`ping 10.0.0.254`	Replies received
Internet access	`ping 8.8.8.8`	Replies received
DNS resolution	`ping google.com`	Name resolves, replies received
Shared folder	File manager in Kali, `/mnt/hgfs/`	Downloads folder visible
Clipboard	Copy text between host and Kali	Text pastes both ways

8.2 If Kali has no Internet

1. Confirm VMnet8 is set to NAT with subnet 10.0.0.0/24 in the Virtual Network Editor.
2. Confirm the VM network adapter is connected to VMnet8.
3. Make sure no other VM on the same network uses 10.0.0.2.
4. Restart the connection and the virtual machine:

```
sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

5. Restart the virtual machines and the host operating system.

9. Conclusion

The lab runs Kali Linux in VMware Workstation Pro on an isolated 10.0.0.0/24 NAT network (VMnet8). The screenshots confirm NAT networking, working Internet and DNS access, and clipboard and drag/drop sharing. Still open against the task: setting the Kali address to 10.0.0.2/24, adding the Downloads shared folder, and completing the snapshot. Phase 2 can then add Windows and Android target machines to the same network.